Sanithu D. Heengama

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EDUCATION

Brown University Providence, RI

Sc.B. in Computer Engineering (Hons) *GPA: 3.8/4.0 Expected May 2027*

Thesis: Precision-Recall Controlled Surface Reconstruction by Curve-Based Lofting and Occlusion Reasoning. Advisor: Prof. Benjamin B. Kimia, Prof. Adriana Schulz. Prof. Leo Kozachkov

3D from Multi-View and Sensors, Brown University (LEMS) — *Advisor: Prof. Benjamin Kimia, Chiang-Heng Chien (Ph.D. Candidate)* Aug 2025 – Present

- Build a multi-view 3D reconstruction pipeline: camera calibration, epipolar geometry, essential/fundamental matrix estimation, triangulation, and optimization for pose recovery.

Musculoskeletal Joint & Implant Tracking, *Brown University Health* — *Orthopaedic Research Lab | Mentor: Prof. J.J. Trey Crisco III* July 2026 – Present

- Extended open-source 3D Slicer tools for markerless skeletal tracking from biplane imaging, applying multi-view reconstruction to a real clinical pose-estimation system.

Radiance Fields for 3D Computer Vision and AI, Brown University — *Advisors: Prof. Srinath Sridhar, Angela Xing (Ph.D. Student)* May 2025 – Dec 2025

- Implemented deformable 3D Gaussian splatting reconstruction in PyTorch for ground-truth particle position/velocity; built a visuotactile manipulation dataset with a custom UMI gripper.

AI Physics Game, Brown University — *Advisor: Prof. Rick Gaitskell* Sept 2025 – Jan 2026

- Designed reinforcement-learning agents and a multi-agent gravitational-interaction environment for stable orbital control.

ADDITIONAL RESEARCH & WORK EXPERIENCE

- Esophageal Manometry Redesign, Brown University & RISD (Sept 2025 – May 2026) — co-led embedded electronics/software for a patient-facing diagnostic device.
- Reaction-Diffusion Pattern Formation Theory, Applied Math DRP, Brown University (Feb – May 2025) — modeled pattern formation; presented at DRP 2025 Workshop.
- Engineering Research Intern, Dexwet Holding Corporation, Vienna, Austria (June – Aug 2023) — developed 9 IoT air-quality prototypes (CAD, MATLAB).

PROJECTS

- Two-Stage CMOS Op-Amp (Cadence Virtuoso, 0.5 µm) — 86.9 dB gain, 2.4 MHz GBW, 97.9° phase margin, 1.0 V swing, 1.14 mW.
- Pipelined RISC-V Processor (Verilog, FPGA) — designed, verified, and timing-optimized; synthesized and tested on FPGA.
- Tendon-Driven Dexterous Hand, Personal Project — built/calibrated a 5-finger open-source hand (ESP32-S3, 7 servos, Kevlar tendons) for stable grasping.
- 16-Camera Multi-View Capture Drone, DoD-Funded — developing a synchronized multi-camera platform for 3D reconstruction research.

PUBLICATIONS

- Heengama, S. (2026). Word-Specific Repetition Priming Sustains Character-Level Language Intrusion in Quantized Reasoning Models. https://doi.org/10.5281/zenodo.22108468
- Heengama, S., Richards, J., & Chong, I. (2026). Humanness without loss of structure: Ensembling ProteinMPNN with AbLang in the RFdiffusion antibody pipeline. (Manuscript)
- Heengama, S., Chien, C.-H., Schulz, A., & Kimia, B.B. Reproducible, Precision-Recall Controlled Surface Reconstruction by Curve-Based Lofting and Occlusion Reasoning. (Manuscript in preparation)
- Heengama, S. "A Cost-Effective IoT-Based Intelligent Indoor Air Quality Monitor." Journal of Emerging Investigators, JEI-23-024R4 (first author, 2023). doi.org/10.59720/23-024

AWARDS, TEACHING & TRAINING

- 3× UTRA/SPRINT Fellowship; HKIMO Global Final — World Star & Bronze Medal; 2× International Scholar w/ Honors; 2× Pearson Edexcel Excellence Award.
- Undergraduate TA, ENGN 40 Dynamics & Vibrations and ENGN 32 Intro to Engineering: Design, Brown University.
- LPS Qubit Collaboratory (LQC) Quantum Computing Program — circuits, algorithms, error correction, qubit technologies (Summer 2025).

SKILLS

Languages: Python, C++, C, Java, MATLAB, Verilog

ML / CV: PyTorch, scikit-learn, NumPy, SciPy, CUDA, 3D Reconstruction, Gaussian Splatting

Hardware / EE: ROS2, Cadence, Fusion 360, SolidWorks